Topic 5: Wall Street Warriors[1]

Introduction

In a small town far away, a young man dreamed of one day making it big in finance. He didn't have much to his name, but what he lacked in resources, he made up for in determination and a keen intellect. Day and night, he studied the markets, pored over financial reports, and learned from the masters of finance.

Years went by, and the young man's efforts paid off in spades. He became a billionaire, with a reputation as one of the savviest investors in the world. But he knew that his success was not just his own - it was also a product of the opportunities he had been given, the mentors who had guided him, and the lessons he had learned along the way.

Determined to pay it forward, he set out to create a platform that would enable aspiring finance professionals, regardless of their backgrounds, to learn about finance and investing while potentially earning significant returns. The idea of a stock trading competition was the perfect way to achieve this.

To bring his vision to life, he assembled a team of software engineers to develop the platform. As one of the software engineers, you were tasked with creating the software according to the requirements. The platform allowed young people to compete against each other in real-time using virtual funds to buy and sell stocks, while learning valuable lessons about finance and investing.

[1] Warning: The guidelines in this project are only for a basic simulated version of the real world stock market. Trading in the real stock market is more like hitting a bullseye while blindfolded and spinning in circles. We do not endorse or encourage real world stock trading and you should not ignore the risk in the stock market. Always do your research and consult with a licensed financial advisor before making any investment decisions. So proceed with caution and happy learning (not trading)!

Beginner's Guide to Buying Stocks

1. Create a trading account: Sign up for a trading account with the brokerage firm running the trading competition.
2. Log in to the app: Use your account credentials to log in to the trading competition app.
3. Search for stocks: Look for the list of stocks available for trading in the competition.
4. View stock information: Click on a stock to see more details about the company, its finances, and performance.
5. Select the stock: Choose a stock that you want to buy.
6. Learn more: Read up on the company to help you make a good investment decision. You may use the site https://www.malaysiastock.biz/Corporate-Infomation.aspx?securityCode=*[insert company trading code here, e.g. 1015]* for this project.
7. Buy the stock: Enter the *price* and *number of shares* you want to buy. The trading competition app has rules on the minimum number of shares that you can buy. Keep in mind that the minimum buy is usually one lot, which consists of 100 shares.
8. Place the order: Once you've entered the details, click on the "buy" button to place the order.
9. Order status: Wait for a notification that your order has been matched with a seller. If the order is successful, you'll be notified. If not, you'll be told that your order was unsuccessful.

In trading, you place an order to buy or sell a specific stock. This order represents your request to exchange a certain amount of money for a position in the stock. A trade occurs when your order is matched with an opposite order from another trader. For instance, if you wish to purchase a stock at RM 1000, you must wait until someone else is selling at that same price to execute a trade.

Once the trade is completed, you will *enter* into a position where you hold the stock and its value will fluctuate with the market, or in other words you have an *open position* on the stock. The position will remain open until you decide to *exit* it by listing the stock at a specific price and waiting for someone else to buy it. This action will create another trade and close your existing position.

Competition Rules

1. Duration: The competition will run for six weeks.
2. Initial funds: RM50,000/account
3. Tradable assets: Participants can trade stocks listed on Bursa Malaysia. You can view the list of stocks here: https://www.malaysiastock.biz/Listed-Companies.aspx.
4. Trading hours: Participants can place trades anytime during the trading week (Monday to Friday), during regular market hours (9:00 AM - 12:30 PM and 2:30 PM - 5:00 PM MST).
5. Trading
    - Initial trading period (first three days): During this period, participants can buy stocks without any restrictions on the number of shares they can purchase. They can use their initial funds to buy stocks at the current market price.
    - Automatic matching: When participants place sell orders at a lower price, the system should automatically match them with buy orders at the same price. This ensures that participants can always buy stocks when they are available at a lower price.
    - After the initial trading period: After the first three days, implement the 500-lot rule. Each participant can only buy up to 500 shares of each stock per order on a first-come, first-served basis.
    - Buying from the 500-lot pool: If no participants are selling a particular stock, the system will automatically allocate shares from the 500-lot pool for each stock. This ensures that participants can always buy stocks at the current market price.
    - Replenishing the 500-lot pool: You can choose to periodically replenish the 500-lot pool to maintain liquidity in the market. This could be done daily, weekly, or at any other interval that you deem appropriate for the competition.

6. Trading restrictions:
    - Participants are not allowed to trade on margin[2] or short sell a stock[3].
    - Participants' account balance must be less than 50% by 5:00 PM daily to encourage active trading. If their balance is greater than or equal to 50%, they will be disqualified from the competition.
7. Scoring system: Participants will earn points based on their profit and loss (P&L) during the competition period. The formula for calculating points is as follows: Points = (P&L / Starting Account Balance) x 100.
8. Leaderboard: The app will display a leaderboard that shows the top 10 participants ranked by their total points earned during the competition period.
9. Notifications: Participants will receive notifications when they enter or exit a position, and when their P&L crosses certain thresholds (e.g., RM1000 profit or RM500 loss).
10. Analytics and reporting: The app provides participants with access to a dashboard where they can view their P&L, account balance, open positions, and trade history. Participants can also generate reports that show their performance metrics during the competition period.

[2] Trade on margin: using borrowed funds for trading
[3] Short selling: borrowing shares, selling them, and then buying them back at a later time to profit from a price decline

It's always helpful to have some existing code to work with, even if it's just a starting point. However, it's important to keep in mind that the code is just a guide, and modifications or adaptations may be necessary to meet the specific requirements of the project. Let's work together in a team to build an exceptional trading platform that exceeds all expectations! (Github Code)

Basic Requirement (12 marks)

1. User authentication (2 marks)
   Allow users to create an account and log in to the app securely. It's crucial to encrypt their credentials to prevent unauthorised access. After all, losing all your hard-earned money to a stranger is not exactly the ideal outcome.

2. Real-time stock data (2 marks)
   To ensure that your users are always up-to-date with the latest stock prices and financial information, it is crucial to ulitize an API (application programming interface) that provides real-time access to this data. By leveraging an API, you can update the stock data at regular intervals (every 5 minutes). If you're searching for reliable and comprehensive stock market data, I recommend exploring the article "Top 8 Best Stock Market APIs to Use in 2023". This informative resource delves into various APIs that can meet your needs. Here are a few options to consider:
      - Twelve Data (Twelvedata)
      - Polygon API (Polygon)
      - Yahoo Finance API (RapidAPI for Yahoo Finance)
      - Alpha Vantage (Alpha Vantage)

   For some reason, you cannot implement any of the aforementioned APIs, an alternative option is to utilise the Wall Street Warriors API, developed by UM TEAM. In order to obtain the API key, kindly send us an email with the subject title "**API Key for Group X Occ X WIA1002 UM**". But wait, there's an exciting twist! If you're in the mood for a thrilling adventure, you can take it to the next level by forging your own path. How, you ask? By creating a web scraper that ingeniously extracts stock data from the vibrant Bursa Malaysia.

If you are unfamiliar with APIs, you can refer to the video (APIs for Beginners 2023) to gain an understanding of the concept. Essentially, APIs are used when we need to request some information from the server programmatically. To incorporate real-time stock data into your application, you can utilise the Spring Boot framework (How to create an REST API in Spring boot using Java) to call the chosen API and retrieve the stock data. You are required to use priority queue to store and retrieve real-time stock data effectively.

3. Trading functionality (2 marks)
    - Enable users to place buy and sell orders for a specific stock, including the stock symbol, quantity and expected buying or selling price.
    - Display a suggested price based on the stock symbol to assist users in making informed decisions
    - Execute trades only if the price falls within an acceptable range (1%) of the current market price. For instance, if the market price is RM1000, users should be able to place orders between RM990 - RM1010.
    - Allows users to cancel their pending orders based on the longest time length or highest amount of money
    - Users must maintain a non-negative balance in their accounts before the market closes.

4. Stock search (1 mark)

    Allow users to quickly search for and view information about stocks available for trading. Utilise string matching algorithms like KMP or Boyer-Moore to optimise the search by name or ticker symbol.

5. Trading dashboard (1 mark)

   Provide a user dashboard displaying their account balance, current P&L or points, open positions, and trade history to enhance their trading experience and allow for easy monitoring of their progress. Users should be able to sort their trade history based on price and placement time.

6. Leaderboard (1 mark)

   Display a leaderboard that showcases the top 10 users ranked by their cumulative points earned during the designated competition period. Implement sorting algorithms to enable ranking functionality, motivating users to enhance their performance and earn a spot on the leaderboard.

7. Notifications (1 mark)

   Send users notifications when they enter or exit a position or when their P&L crosses certain thresholds that the user has set to improve their trading experience and provide timely updates. Here are some general steps to implement the notification function:

   - Create a Notification class that includes the necessary fields.
   - Utilise a third-party library such as JavaMail to send notifications to users.
   - Implement methods to handle user settings for notifications, such as configuring the P&L thresholds or enabling/disabling notifications.
   - Use a timer or scheduling framework like Quartz to periodically check for threshold crossing.
   - Implement logic to send the appropriate notifications based on the user's settings and trading activity.

8. Reporting (1 mark)

   Users should be able to generate reports that present their performance metrics during the competition period. The reports may include commonly used tables, graphs or charts for financial data analysis. These reports should be available in standard formats like .txt, .csv, or .pdf, which will enable users to evaluate their trading strategies and improve their performance.

9. Admin panel (1 mark)
   Provide an admin panel for centralised control of the platform. The admin should be able to disqualify participants who violate the competition rules to ensure a fair trading environment.

Additional challenges (4 marks)

1. Graphical User Interface (GUI): The app's interface should be intuitive and user-friendly to ensure that participants can easily navigate and use the app's features.

2. Centralised database: It is recommended to store all transactions in a centralised database, as it will be easier to manage and keep data in sync. You can use an online database service or a server to store transaction results, but during the code execution, it would be best to store the information as a data structure in the app for quick access.

3. Backtesting: Enable users to backtest their trading strategies against historical data to evaluate how their strategies would have performed in the past. This feature can help users refine their strategies and make more informed trading decisions in the future.

4. Auto fraud detection: The admin panel should automatically flag questionable transactions related to stock fraud, including short-selling, to notify admin members. It is impossible for admins to manually monitor each trade occurring in the system, and hence, automatic fraud detection is crucial.

5. Social Trading: Implement the functionality to enable users to follow and copy trades of other successful traders on the platform.

6. Integration with external APIs: Enhance the app's functionality by integrating with external APIs to provide users with comprehensive financial information. APIs such as news APIs, sentiment analysis APIs, and economic indicators APIs can be integrated.

7. More accurate simulation: To simulate a more accurate trading environment, conduct thorough research and apply financial knowledge in the project.

This assignment requires you to meet all the basic requirements listed. If you miss any of the competition rules, it's game over for you - so pay attention! But hey, don't just stop there. Want to impress your demonstrator and lecturer? Add some optional features to make your app stand out from the rest. Show them what you're made of! Don't be afraid to get creative and explore new functionalities. Who knows, you might even discover something amazing that will make your app the talk of the town. So go ahead, get coding, and let your imagination run wild!

Enquiries

Any questions regarding to this group assignment topic, please contact:
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